AI Image Generator

11. Deployment and Docker

Version 2.1
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Platform: NVIDIA GeForce RTX 3090, Windows 11, Python 3.12

Z-Image Turbo, Stable Diffusion XL and its best fine-tunes, Qwen-Image, FLUX.1, LoRAs and Real-ESRGAN, served locally with Gradio in the Frazile AI studio UI, with per-image GPU telemetry recorded in SQLite and an Admin panel that manages models, LoRAs and downloads and shows live system statistics.

11. Deployment and Docker

11.1 Image

The Dockerfile builds on `nvidia/cuda:12.6.3-cudnn-runtime-ubuntu24.04`, installs Python 3.12, PyTorch and torchvision for CUDA 12.6, then requirements.txt, and copies the Python sources. The Hugging Face cache is pointed at /cache/huggingface, Gradio is told to bind 0.0.0.0:7860, and FRAZILE_IN_DOCKER=1 stops the app from trying to open a browser. A health check polls the root page every 30 s after a 3-minute start grace.

11.2 Compose service

Setting	Value	Why
ports	7860:7860	The studio on http://localhost:7860
volumes	./outputs, ./models, ./generations.db, hf-cache	Images, your checkpoints and LoRAs, the history database, and the downloaded models all live outside the container
environment	HF_TOKEN	Optional; lets the gated FLUX repository download
deploy.resources.reservations.devices	driver nvidia, count all, capabilities gpu	GPU passthrough
shm_size	2 GB	Shared memory for data loaders
restart	unless-stopped	Comes back after a reboot of the host

11.3 Host requirements

Host	Needs
Windows 11	WSL 2 (`wsl --install`, reboot), Docker Desktop with the WSL 2 backend, the normal NVIDIA Windows driver. GPU support is built into Docker Desktop; no extra toolkit.
Linux	Docker Engine, the NVIDIA driver, nvidia-container-toolkit configured as a Docker runtime.

The development machine has neither WSL nor Docker installed, so the image has been written and reviewed but not built or run there. The first build on any machine downloads several GB of Python packages and should be run on a fast connection.

11.4 Commands

```
docker compose build           # build the image
docker compose up              # run in the foreground
docker compose up -d           # run detached
docker compose logs -f         # follow the app log
docker compose down            # stop; keeps the hf-cache volume
docker compose down -v         # stop and delete cached models
```

11.5 Differences inside the container

- Paths in the database are container paths (/app/outputs/...). Mount the same folders when moving between native and container use, or the History tab shows "not found" for the other side's files.
- The Admin panel's CPU, memory and network sections describe the container's view; GPU and power still come from the host driver through nvidia-smi.
- On Linux hosts create an empty generations.db before the first `up`, otherwise Docker creates a directory in its place.
- There is still no authentication. Publishing port 7860 beyond localhost exposes the studio to the network.